

NOAC BRIEFING

Available Reference :

- DOAC Guideline in AF & VTE (updated July 2024)

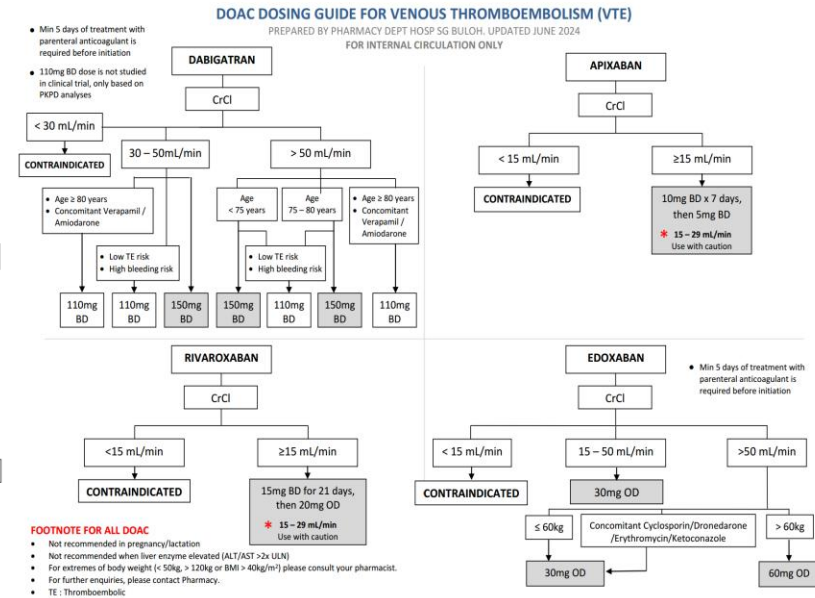
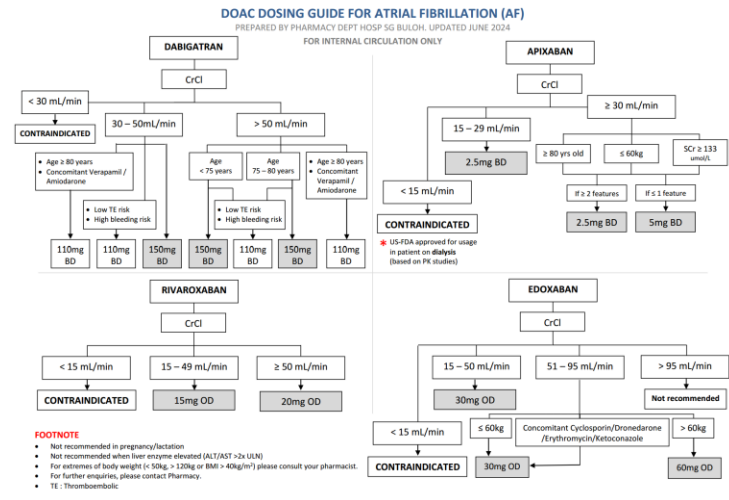
DOAC Guideline in AF & VTE				
Prepared by PHIC HqHq. For internal circulation only. To consult physician if needed.				
CHECKLIST	DABIGATRAN	APIXABAN	RIVAROXABAN	EDOxabAN
INDICATION: STROKE PREVENTION IN AF (SPAF)	For SPAF: Usual dose ¹⁻⁴ 150mg BD Dosing Adjustment ¹⁻⁴ • 110mg BD if fulfill ANY of the following criteria: i) Patients aged ≥80 years ii) Concomitant use of Verapamil / Amiodarone *Please refer to bottom of the table for further dosing consideration. For VTE: Treatment / Prevention ¹⁻⁴ PARENTERAL ANTICOAGULANT for at least 5 days THEN 150mg BD OR 110mg BD if fulfill any of the following risk factors i) Patients aged ≥80 years ii) Concomitant use of Verapamil / Amiodarone iii) High bleeding risk AND CrCl: 30-50 ml/min *Please refer to bottom of the table for further dosing consideration	For SPAF: Usual dose ¹ 5mg BD Dosing Adjustment ¹ • 2.5mg BD if fulfill TWO of the following criteria: i) Patients aged ≥80 years ii) Body weight ≤60kg iii) Serum Creatinine ≥133 umol/L • 2.5mg BD if CrCl 15-29ml/min For VTE: Treatment ¹ 10mg BD for 7 days THEN 5mg BD Prevention of Recurrent VTE ¹ 2.5mg BD (after at least 6 months of therapeutic dose of anticoagulant)	For SPAF: Usual Dose ¹⁻⁴ 20mg OD Dosing Adjustment ¹ • 15 mg OD if CrCl 15-49 ml/min • 10mg OD if fulfill ANY of the following criteria: i) CrCl 15 – 50ml/min ii) Body weight ≤60kg iii) Concomitant use of cyclosporin, dronedarsone, erythromycin, ketoconazole For VTE: Treatment / Prevention ¹⁻⁴ PARENTERAL ANTICOAGULANT for at least 5 days THEN 20mg once daily OR 30mg OD if fulfill any of the following risk factors i) CrCl 15 – 50ml/min ii) Body weight ≤60kg iii) Concomitant use of cyclosporin, dronedarsone, erythromycin, ketoconazole	For SPAF: Usual Dose ¹⁻⁴ 60mg OD Dosing Adjustment ¹⁻⁴ • 30mg OD if fulfill ANY of the following criteria: i) CrCl 15 – 50ml/min ii) Body weight ≤60kg iii) Concomitant use of cyclosporin, dronedarsone, erythromycin, ketoconazole For VTE: Treatment / Prevention ¹⁻⁴ PARENTERAL ANTICOAGULANT for at least 5 days THEN 60mg OD OR 30mg OD if fulfill any of the following risk factors i) CrCl 15 – 50ml/min ii) Body weight ≤60kg iii) Concomitant use of cyclosporin, dronedarsone, erythromycin, ketoconazole
INDICATION: VTE	*Not applicable for prevention of VTE in patient undergoing hip or knee replacement surgery Duration for treatment of least 1 month			
RENAL FUNCTION	<30 ml/min: Contraindicated ¹⁻⁴	15 – 29 ml/min : Use with caution ¹ <15 ml/min : Contraindicated ⁴	15 – 29 ml/min : Use with caution ¹⁻⁴ <15 ml/min : Contraindicated ⁴	<15 ml/min: Contraindicated ¹⁻⁴ >95 ml/min: Not recommended for SPAF due to increased risk of ischemic stroke ¹⁻⁴
* FOR DABIGATRAN: May consider adjustment to 110mg BD based on individual thrombotic and bleeding risk assessment for the following group: i) Age 75-80 years, ii) Moderate renal impairment (30 – 50ml/min), iii) Is of gastric, esophageal or gastroesophageal reflux, iv) Patients with increased risk of bleeding * FOR APIXABAN: IS FDA approved the use of Apixaban 5mg BD in ESRF on regular dialysis based on PK studies. ¹⁻⁴ Reduce 1. Apixaban to 2.5mg BD if ESRF patient on regular dialysis presented with one of the following patient characteristics, i) Age ≥80 years, ii) Body weight ≤60kg				

- Anticoagulant Conversion Chart (updated July 2024)

ANTICOAGULANT CONVERSION CHART						
Alternative for those cannot be found, use Warfarin & DOAC to fill						
From	Warfarin	Dabigatran	Rivaroxaban	Apixaban	Edoxaban	Apixaban
LAMNUS						
From Dabigatran	Overlapped at least 1 day DOAC, until 2 consecutive days with INR in therapeutic range. ^{1,2}	Start DOAC at the next scheduled dose of LAMNUS/Fromdabigatran. ^{1,4,5,6}				
Continuous UFH		Start DOAC at time of discontinuation of continuous UFH. ^{1,4}		Discontinue the UFH infusion; Start Edoxaban 4 hours after discontinuation of UFH infusion. ⁴		Stop continuous UFH; start LAMNUS/Fromdabigatran within 3 hour. ²
Warfarin		Stop Warfarin; start Dabigatran when INR < 2.0. ^{1,5,6}	Stop Warfarin; start Rivaroxaban when INR < 3.0. ^{1,5} DOAC or PE. ⁷ Stop Warfarin; start Rivaroxaban when INR < 2.5.	Stop Warfarin; start Apixaban when INR < 2.0. ^{1,5,6}	Stop Warfarin; start Edoxaban when INR < 2.5. ⁴	Stop Warfarin; start parenteral anticoagulant when INR < 2.0. ²
Dabigatran	Option 1: Overlapping ⁴ Start Warfarin; then DOAC 3.0-5.0mg/day; Stop Dabigatran 3 days later DOAC 30.0-50.0mg/day; Stop Dabigatran 3 days later DOAC 30.0-50.0mg/day; Stop Dabigatran 1 day later DOAC > 50.0mg/day; Dabigatran not recommended. ⁷ Option 2: Bridging ⁴ Start Warfarin; then Warfarin with parenteral agent at most scheduled dose of Dabigatran. Event, parenteral agent until INR in therapeutic range. Option 3: Overlapping ^{1,4} Start Warfarin; then Warfarin with parenteral agent at most scheduled dose of DOAC. Event, parenteral agent until INR in therapeutic range.					DOAC > 50 mg/day ¹ Start Dabigatran; start parenteral agent 1.0 hours after last Dabigatran. DOAC < 50 mg/day ⁴ Start Dabigatran; start parenteral agent 24 hours after last Dabigatran. (Event, Dabigatran contraindicated).
Rivaroxaban/ Apixaban	Option 1: Overlapping ^{1,4} Start Warfarin; then Warfarin with parenteral agent at most scheduled dose of Rivaroxaban. Event, parenteral agent until INR in therapeutic range. Option 2: Bridging ^{1,4} Stop DOAC; start Warfarin with parenteral agent at most scheduled dose of Rivaroxaban. Event, parenteral agent until INR in therapeutic range.	Stop current DOAC; start new DOAC at the next scheduled dose of previous DOAC. ^{1,4}				
Edoxaban	Option 1: Overlapping ^{1,4} INR to be measured before Edoxaban is administered. Start Warfarin; then Warfarin with parenteral agent at most scheduled dose of Edoxaban by 100% for up to 14 days and stop when INR in therapeutic range. Option 2: Bridging ⁴ Stop Edoxaban; start Warfarin with parenteral agent at most scheduled dose of Edoxaban. Event, parenteral agent until INR in therapeutic range.					Stop Rivaroxaban/Apixaban/Edoxaban; start parenteral agent at the next scheduled dose of Rivaroxaban/Apixaban/Edoxaban. ^{1,4}
Only for internal circulation. For further enquiries, kindly contact our ELSU. Revised July 2024.						
References: 1. CHE Prevention and Treatment of Venous Thromboembolism. 2023. 1-106. 2. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 3. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 4. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 5. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 6. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 7. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 8. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 9. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016). 10. LAMNUS® (Dabigatran Etexilate). P. 17. Pfizer. (London: Pfizer, 2016).						

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CHECKLIST	DABIGATRAN	APIXABAN	RIVAROXABAN	EDOxabAN
HEPATIC FUNCTION	Child Pugh A: No dosage adjustment. Child Pugh B: Use with caution. Child Pugh C: Contraindicated. ¹⁻⁴ Not recommended when elevated liver enzymes >2x ULN. ¹⁻⁴	Child Pugh A: No dosage adjustment. Child Pugh B: Use with caution. Child Pugh C: Contraindicated. Not recommended when elevated liver enzymes >2x ULN. ¹⁻⁴	Child Pugh A: No dosage adjustment. Child Pugh B: Contraindicated. Child Pugh C: Contraindicated. Not recommended when elevated liver enzymes >2x ULN & 3x ULN. ¹⁻⁴	Child Pugh A: No dosage adjustment. Child Pugh B: Contraindicated. Child Pugh C: Contraindicated. Not recommended when elevated liver enzymes >2x ULN & 3x ULN. ¹⁻⁴
PREGNANCY & LACTATION	Not recommended. ¹⁻⁴	Not recommended. ¹⁻⁴	Not recommended. ¹⁻⁴	Not recommended. ¹⁻⁴
WEIGHT CONSIDERATION	• Wt <50kg: Use with caution. ¹⁻⁴ • BMI >40kg/m ² or Wt >120 kg: Avoid use. Suggest to use alternative. ¹⁻⁴ * Use body weight. * Avoid use. Suggest to use alternative. ¹⁻⁴ * Caution: 1 drug exposure	• Wt <50kg (Use with caution) For SPAF: Adjust dose if other criteria are met. For VTE : No dosage adjustment. ¹⁻⁴ • BMI >40kg/m ² or Wt >120 kg: Use with caution. Consult physician. ¹⁻⁴	• Wt <50kg: Use with caution. ^{1,4,5,6} • BMI >40kg/m ² or Wt >120 kg: No dosage adjustment. ¹⁻⁴	• Wt <50kg: 30mg OD ¹⁻⁴ • BMI >40kg/m ² or Wt >120 kg: For SPAF: No dosage adjustment. For VTE : Avoid use.
ADMINISTRATION	Swallow whole. Do NOT break, chew or open capsules. ¹⁻⁴ For VTE: Treatment / Prevention ¹⁻⁴ Oral pellet : 3 months to <12 years old ¹⁻⁴ (Not available in Malaysia) Capsule : 18 to <18 years old ¹⁻⁴ The dosage forms are NOT interchangeable on a mg-for-mg basis due to differences in bioavailability. No safety & efficacy studies. IV Idarucizumab 5g STAT • Administer as two separate 2.5g doses with NO >15 mins apart. Alternative if Idarucizumab is not available: IV Octaplex 50 units/kg STAT (Max: 4000 units)	No safety & efficacy studies. Recommended. ¹⁻⁴ LAMNUS more preferred for gastrointestinal/hepatobiliary related cancer. ¹⁻⁴ IV Andexanet alfa (Not available in HqHq) Alternative: IV Octaplex 25-50 units/kg STAT OR IV Octaplex 2000 units STAT	Swallow whole or crush the tablet and suspend in 50ml of water / DS. To be taken with food. ¹⁻⁴ For VTE: Treatment / Prevention ¹⁻⁴ From birth to <18 years old: Oral suspension : ≥2.6 kg (Not available in Malaysia) Tablet : ≥30 kg Recommended. ¹⁻⁴ LAMNUS more preferred for gastrointestinal/hepatobiliary related cancer. ¹⁻⁴ IV Andexanet alfa (Not available in HqHq) Alternative: IV Octaplex 25-50 units/kg STAT OR IV Octaplex 2000 units STAT	Swallow whole or crush the tablet and suspend in small amount of water. Stable up to 4 hours. ¹⁻⁴ No safety & efficacy studies. Recommended. ¹⁻⁴ LAMNUS more preferred for gastrointestinal/hepatobiliary related cancer. ¹⁻⁴ IV Andexanet alfa (Not available in HqHq) Alternative: IV Octaplex 25-50 units/kg STAT OR IV Octaplex 2000 units STAT
PHYSICAL APPEARANCE	Pradaxa hard cap 150mg Pradaxa hard cap 110mg	Elquis FC Tab 5mg Elquis FC Tab 2.5mg	Karelio FC Tab 20mg Karelio FC Tab 15mg	Litena FC Tab 60mg Litena FC Tab 30mg

- **DOAC Dosing Guide for VTE & AF (updated July 2024)**



- **CME DOAC : Farmasi > Kaunseling > Validasi PRP > Counselling Slides**

Available DOAC

- Dabigatran 110mg & 150mg
- Apixaban 2.5mg & 5mg
- Rivaroxaban 15mg & 20mg
- Edoxaban 30mg & 60mg

**** Kuota / Pesara / SELF PURCHASED**

DOAC Screening

- Must be done by PHARMACIST (PF)
- Use DOAC Cop / DOAC Calculator
- Refer DOAC Guideline in AF & VTE, DOAC Dosing Guide for VTE & AF Newsletter
- Indication, Choice of DOAC, Method of Supply, Contraindications, Age, RTF, BW, RP, LFT, Hb, Last Parenteral Dose, Interactions
- DAPT / Triple Tx Duration
- Starting Date DOAC (remind Dr to OFF Parenteral Anticoagulant once switching to DOAC), confirm duration of treatment with Dr
- Special Consideration: OFF Label Use in Adolescence
- Interaction Checker: UptoDate

DOAC Counselling Notes

- Use DOAC Counselling template
- Must include: last parenteral dose, remind Dr / SN to off parenteral anticoagulant once switching to DOAC
- Ensure patient bought the correct DOAC / check the manual Rx given by Dr
- Record any intervention / discussion with Dr
- Key in correct date for VTE
- Include SCREEND BY PF _____

GENERAL INFO

- DOAC is HIGH ALERT MEDICATION !
- Prescribing Error > new dr, complicated regimen / dosing
- EXTRA steps when screen for DOAC
- MAKE SURE DOAC order in system : easy to trace